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NOMOLOGICAL ALGEBRA.

By Elworthy, of the White Tower Society.

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IV.

NOMOLOGICAL ALGEBRA.

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Communicated to the White Tower Society, C.R. 1061.

Severin's Sixth Question asks: "Can the lines of law be written in a strict reckoning?" When two law-mages speak of a line, he said, they must often gesture at length, and when they have done they may not mean the same line. More than two hundred years have passed, and the case is not much altered: the same line is written in one manner and another in the manuscripts of those who have grasped it; and the craftsmen, in building their arrays and wards, rely upon the rules of experience in Pendleton's Laws of the Craftsmen, rules which tell what to do but not why, so that where the book is silent nothing remains but to try. In forty years, upon the many scattered manners of writing left by those before me, I have settled a manner of writing and of reckoning, which I call nomological algebra. I shall show four things in this paper.

First, every elementary line of law is written as a sign; lines are combined in two ways only, by joint application, in which they are applied together, and by successive application, in which they are applied one after the other; these keep a few simple rules, and successive application in general depends upon the order. Secondly, the two lines of the old wards, the warp and the weft, applied in succession, bring forth a third line; only the three together are complete in themselves, and from this it may be explained why the old wards tear where the load is gathered, and the weave that mends it may be derived. Thirdly, whether several lines can be jointly applied may be decided by a second reckoning. Fourthly, when the supply from without, the exchange with what lies about and the dispersal are all entered in the account, lines that keep the whole amount, however they are combined, leave the whole amount the same before and after; and why the ledger of Finch, eighty years ago, still agrees under combination may be shown from this. Lastly I set out three trials.

I. THE SIGNS

The condition of the Web of Laws at a place I call a web-state, and write w. A line of law applied to the web changes one web-state into another, so that a line is an operation upon web-states, and I write it with a capital letter. The result of applying X to the web-state w is written Xw.

Two web-states may be laid together at one place, which I write as adding them, $w + v$; and a web-state may be enlarged, aw being the web-state with its rising and falling enlarged a times. An operation applied to web-states added together acts upon them term by term: $X(w + v) = Xw + Xv$, and $X(aw) = a(Xw)$. These two are general rules, and no line that I have measured has failed of them.

The operation that changes nothing is called the identity operation, and written 1: $1w = w$. When an operation yields nothing at all, not even a web-state, I say that it yields the null, and write 0. The null is not a web-state: the web at rest, holding no mana, is still a web-state, and is not the null. Any operation applied to the null yields the null; the null added to a web-state yields that web-state; and a web-state multiplied by a number yields the null only if the number is nought. The measure in which a line is applied is written by a number: aX is X applied a times over; the measures are common numbers, positive or negative, whole or broken. $-X$ is the reverse of X, whose rising and falling is just contrary to that of X; a negative measure only applies the line the contrary way, and is no negative mana.

II. JOINT AND SUCCESSIVE APPLICATION

When two lines are applied together, I call it joint application, and write $X + Y$; when Y is applied first and X after, I call it successive application, and write XY. Of the web-state obtained, joint application gives the sum of what the two give, and successive application hands what Y gives to X:

$$(X + Y)w = Xw + Yw, \quad (XY)w = X(Yw) \tag{1}$$

(1) defines the two manners of application. From it and the two general rules of Section I the following rules may be derived:

$$X + Y = Y + X, \quad (X + Y) + Z = X + (Y + Z) \tag{2}$$

$$(XY)Z = X(YZ), \quad X(Y + Z) = XY + XZ, \quad (Y + Z)X = YX + ZX \tag{3}$$

$$a(XY) = (aX)Y = X(aY), \quad 1X = X1 = X, \quad X + (-X) = 0 \tag{4}$$

(2) says that joint application does not depend on the order; (3) that successive application may be grouped as one pleases, and is distributed over joint application; (4) that the measures may be moved before or after, that the identity operation changes nothing, and that a line applied jointly with its reverse cancels its rising and falling and yields the null. When these had been derived, I had the law-mages of White Tower test each of them by casting, and not once has one failed.

Wherever a joint application stands in an equation, the lines in it must all be applicable together, at one place and under the same conditions, before either side has a meaning; which lines can be applied together is reckoned apart, in Section V. An equation of successive application alone is under no such restriction.

One thing only is unlike the reckoning of number: successive application in general depends upon the order, and XY and YX are often not the same. The law-mages have long known that two workings applied in the other order give another result; but without a manner of writing they could not say wherein the result differs, or by how much. The difference of the two orders I write

$$[X, Y] = XY - YX \tag{5}$$

and call the difference of reversal. It compares what the two orders give; it does not mean that both are applied at once. When the difference of reversal of two lines is the null, I say that they may be interchanged.

III. THE WARP, THE WEFT AND THE THIRD LINE

The old wards are all woven of two lines, which the craftsmen call the warp and the weft. I write the warp A and the weft B. Of all the work of forty years, the successive application of A and B cost the most labour.

When B is applied first and A after, there arises at that place of the web a kind of twist, whose direction is neither that of the warp nor that of the weft; when A is applied first and B after, the twist is just contrary. This twist can also be applied by itself, and is another line, which I call the bias, and write C. Measured thus by casting, one by one, the successive applications of the three lines make up Table I.

Table I. The successive application of the warp, the weft and the bias (rows applied after, columns applied first)

After \ First	*A (warp)*	*B (weft)*	*C (bias)*
A (warp)	-1	C	$-B$
B (weft)	$-C$	-1	A
C (bias)	B	$-A$	-1

The -1 in the table is the reverse of the identity operation: when the same line is applied twice in succession, the web-state is turned over, its rising and falling just contrary to what it was; applying A twice, w becomes $-w$. And $-w$ is still a web-state, neither the null nor any negative mana.

From Table I, the differences of reversal of the three lines are

$$[A, B] = 2C, \quad [B, C] = 2A, \quad [C, A] = 2B \tag{6}$$

Any two of the lines applied in succession give again a joint application of 1, A, B and C, each in its own measure. Whatever operation can be written $a + bA + cB + dC$ gives, under joint and successive application, what can again be so written, so that this family is complete in itself. The warp and the weft alone make no such family: A and B applied in succession must bring forth C.

From this it may be explained why the old wards tear. An old ward bears its load always in the same order: the load as it falls is taken first by the warp, and the warp hands it on to the weft. When the load is gathered at one place of a ward, the web there has the warp applied first and the weft after, which by Table I is $BA = -C$, and a twist arises contrary to the bias. In an old ward there is no line of bias, and this portion has nowhere to go, but gathers at that place; when enough has gathered, the ward tears there. Many of the wards that Pendleton recorded tore just where the load was gathered. If a line of bias is woven in besides the warp and the weft, the bias bears a twist in either direction, and the portion spreads along the lines of bias over the whole ward. This much is my explanation: Table I says only that the warp and the weft in succession bring forth a portion of the bias; how that portion gathers and how it spreads the reckoning does not contain, and I have conjectured it from the places where Pendleton's wards tore. But it yields a prediction that can be tried, for which see the third trial of Section VII. That the three sets of lines cross at sixty degrees is so that at every place all three cross

equally close; this is the craftsmen's rule of weaving, and is not derived from the algebra.

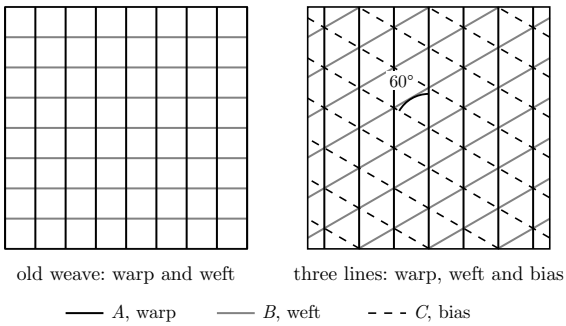


FIG. 1. The old weave of two lines (left) and the weave of three lines (right): a set of lines of bias is added to the warp and the weft, the three sets crossing one another at sixty degrees

IV. THE DUAL AND THE STRENGTH

The dual of an operation X, written $\bar{X}$, does what X does in the contrary way. The dual keeps these rules: the dual of a joint application is the joint application of the duals; the dual of a measure is that same measure, so that the identity operation and its multiples are their own duals; the dual of a successive application is the successive application of the duals in the reverse order, $X\bar{Y} = \bar{Y}\bar{X}$; and the dual of the dual is the operation itself. The dual is not the reverse: the reverse of twice the identity operation is twice its reverse, but its dual is twice the identity operation still.

The lines I have measured keep one general rule: an operation and its dual applied in succession, in whichever order, yield what is never negative; that is, if what they yield merely multiplies some web-state by a number, that number is never less than nought.

The duals of A, B and C are their reverses. For $X = a + bA + cB + dC$ (where a stands for the identity operation in the measure a),

$$\bar{X} = a - bA - cB - dC, \quad \bar{X}X = X\bar{X} = a^2 + b^2 + c^2 + d^2 \tag{7}$$

What is obtained is a number only, multiplying the identity operation, and not less than nought; an operation of this family and its dual applied in succession, in whichever order, give this number. I call it the strength of X, and write it $|X|^2$. When X and Y both belong to this family,

$$|XY|^2 = |X|^2 \ |Y|^2 \tag{8}$$

The proof needs, besides (3), (4) and (7), only the rule that the dual reverses the order: $X\bar{Y}XY = \bar{Y}\bar{X}XY = |X|^2 \ \bar{Y}Y = |X|^2 \ |Y|^2$, the middle step because $|X|^2$ is a number, and may be moved before or after. When two operations are applied in succession, the strength is the product of the two strengths. This the craftsmen can use: how strong a compound line is need not be found by casting it, but can be reckoned from the lines that compose it. It holds only within the family of the warp,

the weft and the bias; raising and lowering are not of this family, and it may not be carried over to them.

V. COMPATIBLE AND INCOMPATIBLE LINES

Joint application requires that two lines can be applied at one place together. Some cannot: no law-mage can make a cold fire, and upon one and the same fire the lines of fire and of cold, however they are woven, admit no construction that satisfies the conditions of both.

First the constructions to be considered are fixed: all that are possible with the given instrument, material, supply and object of the working. The condition of a line is those of these constructions upon which the line can be applied, and the material, the supply and the object must all be written into it. The condition of an elementary line is found by casting, one construction after another; that of a compound line is the product of the conditions of the lines that compose it. I write a condition by a small letter: the condition of fire f, of cold g. That both conditions are to be satisfied I write fg, which is the constructions common to both; a condition said twice is the same condition:

$$ff = f, \quad fg = gf \tag{9}$$

A condition that no construction satisfies I write 0; that within which every construction lies, 1. These 0 and 1 are those of the conditions, and not the null and the identity operation of Section I. "Not satisfying f" I write $1 - f$, and then

$$f(1 - f) = 0 \tag{10}$$

no construction can both satisfy a condition and not satisfy it. The conditions having been written whole, two lines are compatible if, and only if, the product of their conditions is not 0; and so for three or more, the product of all their conditions must not be 0. Whether the supply will serve compatible lines all together is reckoned apart, by the account of Section VI. Fire and cold cannot be jointly applied, because

$$fg = 0 \tag{11}$$

This reckoning is another thing from that of joint and successive application above: that one reckons how lines act upon web-states, this one whether several lines can act together. Its use is that it can reckon what words cannot make plain. For instance, three lines may each be jointly applied with each, and yet not all three together: pq, qr and pr are none of them 0, but pqr is 0. To take a small example: let there be only three constructions, the first, the second and the third; let p be satisfied by the first and second, q by the second and third, and r by the first and third; any two have one construction in common, but all three have none. Pendleton recorded three such overlayings, in which two layers did well and the third spoiled all, and he could only write "not to be laid three deep". I applied the line of each layer in these three by itself upon the same kind of instrument, and found its condition so, without regard to how the overlaying had fared; multiplied by (9) to (11), in all three the products two by two are not 0, and the product of all three is 0. The conditions and the reckoning of each case are kept by the White Tower Society.

VI. THE WHOLE AMOUNT UNCHANGED

Eighty years ago Finch showed that in the turning of the elements, if the elements, the media that store mana, what lies about and the dispersal are entered together, the ledger of mana agrees always within the error of measurement. The combination of lines of law has its ledger also, and it may be written in the algebra.

The mana contained in a web-state is measured by an operation, which I call the amount, and write Q: if $Qw = qw$, the amount of w is said to be the definite number q. The operation that raises one share of mana from the depths of Tiamat up into the web is called raising, and written U; that which sends one share back into the depths, lowering, and written D. Each application of raising adds one share to the amount at that place, and each application of lowering takes one away. The share here is the portion of exchange taken by the usage of reading; how much it is I cannot measure, and know only that it is far below one division of the mana meter; it is not a least quantity that has been proved. Raising and lowering are dual to each other, and by the general rule of Section IV, applied in succession, in whichever order, they yield what is never negative. So far as I can measure, raising-then-lowering and lowering-then-raising do not differ: the one order gives the same as the other.

A line that may be interchanged with the amount, $QX = XQ$, I call a line that keeps the amount: applied to a web-state of definite amount q, it yields, unless it yields the null, a web-state whose amount is still q, since $Q(Xw) = X(Qw) = q(Xw)$. Conversely, a line that leaves unchanged the amount of every web-state of definite amount may be interchanged with Q, since every web-state that I have measured can be taken apart into several such web-states added together. Lines that keep the amount have one property of consequence:

$$QX = XQ, \quad QY = YQ$$
$$\Rightarrow \quad Q(X+Y) = (X+Y)Q, \quad Q(XY) = (XY)Q \tag{12}$$

Lines that keep the amount, however they are applied jointly and successively, give again what keeps it. The proof needs only (1) to (3): $Q(XY) = (QX)Y = (XQ)Y = X(QY) = X(YQ) = (XY)Q$; and so for joint application. A combination must of course be applicable, and must yield something other than the null, before there is any amount for it to keep; (12) says that every combination that can be written keeps the amount, not that every one that can be written can be applied.

Looking at one place of the web alone, many lines do not keep the amount: raising draws mana up from the depths, and dispersal sends it out beyond the bounds. But if the supply from without, the exchange with what lies about and the dispersal are all entered in the account, the account holds. It has five items: first, the amount upon the web at that place; secondly, what the feeding crystal still holds; thirdly, the item of the depths, in which the shares raised from the depths and not yet lowered back are entered as paid out, with the negative sign; fourthly, what has passed into what lies about; fifthly, the item of dispersal, in which what has been dispersed beyond the bounds is entered as it accumulates. When one share

is raised, the web has one share more and the item of the depths one less; when the feeding crystal gives one share, the crystal has one less and the web one more; when one share is dispersed, the web has one less and the item of dispersal one more. The operation that measures the whole of the five items together I write T, to keep it apart from Q, which measures one place alone; the web-state is then taken to hold the other four items also. Every elementary line that I have measured may be interchanged with T. With T in place of Q, (12) holds as before, so that all the arrays and wards combined from these lines, so far as they can be applied, keep this whole amount also. This agrees with the ledger of Finch: when the supply from without, the exchange with what lies about and the dispersal are entered together, the whole mana is the same before and after the combination.

Each elementary line that I have measured I have entered by Finch's rules, and its receipts and payments differ within the error of measurement; the account of each line is kept by the White Tower Society. From this it may be seen also that the ledger still agrees under combination by no chance: so long as every elementary line keeps the whole amount, every combination however intricate keeps it, and there is no need to measure them one by one. Why the elementary lines keep the whole amount I cannot say; Finch said that he could not tell why the ledger always agrees, and here the question has only been carried back to the elementary lines.

VII. THREE TRIALS

A manner of writing is not enough because it is orderly; it must reckon what the casters can measure. I have made three trials.

First, reckoning against casting. After Table I was fixed, I reckoned beforehand by the rules above what two hundred and sixteen combinations should give, and wrote the reckonings on sealed sheets; then I asked twelve law-mages of White Tower to test them by casting. Each combination was cast once by each of two law-mages working apart, so that each of the twelve cast thirty-six times; those casting did not see the reckoning, and the sheets were opened only when the casting was done. The fifty-four combinations of the first kind are new measures and orders beyond Table I, and not the old records from which the table was fixed. A result was counted as agreeing in this way: for successive application, the direction of the rising and falling and the measure of each term; for strength, the number measured; for joint application, whether it could be applied or not; each the same within what the casting and the instruments could distinguish. The results are in Table II; all agree, which is to say that within the precision used no difference was seen. The sheets and the records of casting are kept by the White Tower Society.

Table II. Reckoning tested by casting

What was reckoned	*Number*	*Agreeing*
Order and result of two lines in succession (Table I and (6))	54	54
Result of three or more lines in succession	72	72
Strength of compound lines ((8))	48	48
Whether lines can be jointly applied ((9) to (11))	42	42
Sum	216	216

Secondly, Pendleton's overlayings. The Laws of the Craftsmen records thirty-one overlayings that failed; Pendleton set down how they failed, but not why. By nomological algebra all thirty-one can be derived, as in Table III. They are sorted by their chief cause, each into one kind only: first the product of conditions is looked at, and those in which it is 0 go into the second kind; of the rest, the order is looked at, and those in which the difference of reversal brings forth a portion of a line that no engraving upon the instrument bears, and which, like the portion of bias in the old wards of Section III, has nowhere to go, go into the first kind; the rest are reckoned by the account of Section VI, how much more the web must hold after the combination is applied than before, and so how much must be drawn from the supply, and in all three it was more than the supply could give: these are the third kind. The conditions, the differences of reversal and the amounts needed were all found by casting each line by itself and reckoned from the records of its casting, without regard to the failures. I took besides forty overlayings from the Laws of the Craftsmen, made under like conditions, which did well, and reckoned them in the same manner; not one was reckoned to fail.

TABLE III. The thirty-one failed overlayings of the Laws of the Craftsmen

Cause	*Number*	*Derived by the algebra*
Wrong order of succession, the difference of reversal bringing forth a portion that should not be	19	19
Product of conditions 0 (3 of them compatible two by two, but not all together)	9	9
Amount needed by the combination greater than the supply	3	3
Sum	31	31

Thirdly, the weave of three lines. If the conjecture of Section III is right, a ward of three lines should not tear under a prescribed load, duration and disturbance. This no one had said before, and it can be tried. I asked the craftsmen of White Tower to weave, of one batch of materials and in like frames, twenty-four trial pieces six feet square. Twelve were woven of three lines: the warp and the weft at the common spacing, with a set of lines of bias added. Twelve were woven of two lines in the old manner: six at the common spacing, and six with the warp and the weft drawn closer, until the whole length engraved was the same as in the weave of three, to see whether more lines of the old kind alone would serve as well as the bias. The supply of each piece was given in proportion to the whole length engraved. The working load that the craftsmen allow to the common weave of two lines I count as once, and both kinds were reckoned by it. The trial laid a gathered load upon the centre of each piece by an array, beginning at once the working load and adding a tenth of it evenly in every quarter of an hour, with a disturbance of the same size added at the same place from first to last, until the piece tore, or the load reached three times; at three times it was held for two hours more. The twelve of the old weave all tore: the six at the common spacing between 1.62 and 1.93 times the working load,

the six drawn closer between 1.97 and 2.29; of the twelve of three lines, pressed to three times and held two hours more, not one tore. See Fig. 2.

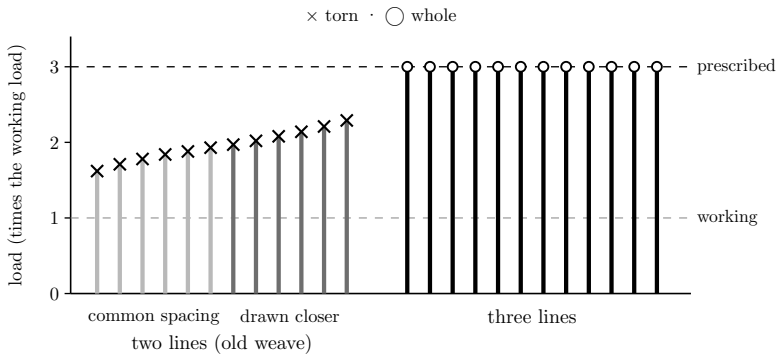


Fig. 2. The twenty-four trial pieces, in multiples of the working load: left, the twelve of the old weave of two lines, the first six at the common spacing and the last six drawn closer, all torn before three times, drawn at the load at which they tore, each six in order of that load; right, the twelve of the weave of three lines, whole at the prescribed three times after two hours more

VIII. WHAT CANNOT BE WRITTEN IN THE RECKONING

Nomological algebra is not all. The law-mages who hold most by what is grasped in ascending the Tower say that what is written in the reckoning is only the shadow of what is grasped; and this I do not wholly deny: the reckoning keeps how lines are added and applied in succession, but not all that is seen at the moment of grasping. For many workings of the witches I find no reckoning that answers; whether they do not go by the Web of Laws at all, or go by lines I have not written, I do not know. Nor can I say why the lines of law keep these rules; I know only that in forty years they have not once failed to keep them.

I write the manner of writing here in the hope that those who grasp a line may henceforth, when they speak of it, write it down and compare, and need no longer gesture at length.